Data Dictionary

This section presents the data dictionary of the WebXR-Enabled Digital Heritage Platform for the NHCP Museo ng Libingan sa Ilalim ng Lupa ng Nagcarlan. Each appendix table identifies the final field name, PostgreSQL data type, principal key or nullability constraint, and functional meaning of the field.

Table A1. Data Dictionary for admin_users

FIELD NAME	DATA TYPE	CONSTRAINTS	DESCRIPTION
id	BIGINT	PK; Identity; Required	Unique identifier for the record.
email	TEXT	Unique; Required	Normalized email address used by the account.
password_hash	TEXT	Required	One-way password hash used to authenticate the account.
is_active	BOOLEAN	Required	Indicates whether the account or category is available for use.
created_at	TIMESTAMPTZ	Required	Date and time when the record was created.
updated_at	TIMESTAMPTZ	Required	Date and time when the record was last updated.

Table A1 stores administrator credentials and account availability for the protected management portal.

Table A2. Data Dictionary for admin_notifications

FIELD NAME	DATA TYPE	CONSTRAINTS	DESCRIPTION
id	BIGINT	PK; Identity; Required	Unique identifier for the record.
recipient_email	TEXT	FK → admin_users.email; Composite unique; Required	Email address of the administrator receiving the notification.
event_type	TEXT	Composite unique; Required	Application event that generated the

			notification or analytics record.
source_id	BIGINT	Composite unique; Required	Identifier of the source record that generated the notification.
title	TEXT	Required	Primary title of the record.
message	TEXT	Required	Short notification message shown to the administrator.
target_url	TEXT	Required	Protected administration page opened from the notification.
read_at	TIMESTAMPTZ	Optional	Date and time when the notification was marked as read.
emailed_at	TIMESTAMPTZ	Optional	Date and time when the notification email was sent.
email_error	TEXT	Required	Last email-delivery error recorded for the notification.
created_at	TIMESTAMPTZ	Required	Date and time when the record was created.

Table A2 stores the persistent notification inbox created for each administrator from visitor and community activity.

Table A3. Data Dictionary for admin_password_reset_requests

FIELD NAME	DATA TYPE	CONSTRAINTS	DESCRIPTION
id	BIGINT	PK; Identity; Required	Unique identifier for the record.
admin_user_id	BIGINT	FK → admin_users.id; Optional	Identifier of the related administrator account.
email_hash	TEXT	Required	HMAC digest of the submitted recovery email address.
request_ip_hash	TEXT	Required	HMAC digest of the client address used for recovery rate limiting.
requested_at	TIMESTAMPTZ	Required	Date and time when the recovery request was submitted.

Records privacy-preserving password-recovery requests for rate limiting and security review.

Table A4. Data Dictionary for admin_password_reset_tokens

FIELD NAME	DATA TYPE	CONSTRAINTS	DESCRIPTION
id	BIGINT	PK; Identity; Required	Unique identifier for the record.
admin_user_id	BIGINT	FK → admin_users.id; Required	Identifier of the related administrator account.
token_hash	TEXT	Unique; Required	One-way hash of the single-use recovery token.
expires_at	TIMESTAMPTZ	Required	Date and time after which the token is no longer valid.
used_at	TIMESTAMPTZ	Optional	Date and time when the token was consumed; null while unused.
created_at	TIMESTAMPTZ	Required	Date and time when the record was created.

Table A4 stores single-use hashed recovery tokens issued to administrator accounts.

Table A5. Data Dictionary for public_users

FIELD NAME	DATA TYPE	CONSTRAINTS	DESCRIPTION
id	BIGINT	PK; Identity; Required	Unique identifier for the record.
email	TEXT	Unique; Required	Normalized email address used by the account.
username	TEXT	Unique; Required	Unique public username used for account identification.
password_hash	TEXT	Required	One-way password hash used to authenticate the account.
full_name	TEXT	Required	Full display name of the public user.
phone_number	TEXT	Required	Contact telephone number supplied by the user.
organization	TEXT	Required	Organization or institution supplied in the user profile.
is_active	BOOLEAN	Required	Indicates whether the account or category is available for use.

last_login_at	TIMESTAMPTZ	Optional	Date and time of the user's most recent successful sign-in.
email_verified_at	TIMESTAMPTZ	Optional	Date and time when the user's email address was verified.
email_verification_token_hash	TEXT	Optional	Hash of the current email-verification token.
email_verification_expires_at	TIMESTAMPTZ	Optional	Expiration date and time of the email-verification token.
email_verification_sent_at	TIMESTAMPTZ	Optional	Date and time when the latest verification message was sent.
email_verification_attempts	INTEGER	Required	Number of failed attempts made against the current verification code.
onboarding_completed_at	TIMESTAMPTZ	Optional	Date and time when the user completed profile onboarding.
website_use_case	TEXT	Required	User-selected reason for using the heritage platform.
profile_roles	TEXT	Required	Serialized list of roles or visitor types selected during onboarding.
location_text	TEXT	Required	Free-text location originally supplied by the user.
country_text	TEXT	Required	Country saved in the user's profile.
province_text	TEXT	Required	Province saved in the user's profile.
city_text	TEXT	Required	City or municipality saved in the user's profile.
auth_provider	TEXT	Required	Authentication method associated with the account.
oauth_subject	TEXT	Unique; Optional	Unique external identity subject supplied by the OAuth provider.
local_password_enabled	BOOLEAN	Required	Indicates whether password-based sign-in is enabled for the account.
password_reset_token_hash	TEXT	Optional	Hash of the user's active password- reset token.

password_reset_expires_at	TIMESTAMPTZ	Optional	Expiration date and time of the user's password-reset token.
security_code_hash	TEXT	Optional	Hash of the temporary account- security verification code.
security_code_expires_at	TIMESTAMPTZ	Optional	Expiration date and time of the temporary security code.
security_code_sent_at	TIMESTAMPTZ	Optional	Date and time when the latest security code was sent.
security_code_attempts	INTEGER	Required	Number of failed attempts made against the current security code.
profile_picture	TEXT	Optional	External image URL or storage object path for the user's profile picture.
theme_preference	TEXT	Required	Preferred interface theme: light or dark.
language_preference	TEXT	Required	Preferred interface language: English or Tagalog.
created_at	TIMESTAMPTZ	Required	Date and time when the record was created.
updated_at	TIMESTAMPTZ	Required	Date and time when the record was last updated.

Table A5 stores public visitor accounts, authentication state, profile information, onboarding data, and interface preferences.

Table A6. Data Dictionary for user_bookmarks

FIELD NAME	DATA TYPE	CONSTRAINTS	DESCRIPTION
id	BIGINT	PK; Identity; Required	Unique identifier for the record.
user_id	BIGINT	FK → public_users.id; Required	Identifier of the related public user.
item_type	TEXT	Required	Module type of the bookmarked content.
item_id	BIGINT	Required	Identifier of the bookmarked content record.

created_at	TIMESTAMP TZ	Required	Date and time when the record was created.

Table A6 links a public user to an online-gallery item saved for later viewing.

Table A7. Data Dictionary for tour_bookings

FIELD NAME	DATA TYPE	CONSTRAINTS	DESCRIPTION
id	BIGINT	PK; Identity; Required	Unique identifier for the record.
user_id	BIGINT	FK → public_users.id; Required	Identifier of the related public user.
display_title	TEXT	Required	User-facing title assigned to the booking.
scheduled_tour	TEXT	Required	Name of the selected museum tour or visit activity.
booking_type	TEXT	Required	Category of the visiting group or requested tour.
visit_date	DATE	Required	Requested date of the museum visit.
rescheduled_visit_date	DATE	Optional	Replacement visit date when the booking is rescheduled.
start_time	TEXT	Required	Requested starting time stored in display format.
end_time	TEXT	Required	Requested ending time stored in display format.
group_size	INTEGER	Required	Total group size entered when the booking was created.
institution_name	TEXT	Required	School, organization, or institution represented by the group.
contact_person	TEXT	Required	Primary contact person for the booking.
contact_number	TEXT	Required	Telephone number of the booking contact.

tour_agency	TEXT	Required	Tour agency associated with the visit, when applicable.
male_visitors	INTEGER	Required	Number of male visitors included in the booking.
female_visitors	INTEGER	Required	Number of female visitors included in the booking.
pwd_visitors	INTEGER	Required	Number of persons with disabilities included in the booking.
total_visitors	INTEGER	Required	Final total number of visitors included in the booking.
purpose	TEXT	Required	Purpose or objective stated for the museum visit.
notes	TEXT	Required	Additional information supplied with the booking.
location	TEXT	Required	Location associated with the booking or content item.
status	TEXT	Required	Current workflow or publication state of the record.
admin_notes	TEXT	Required	Internal notes recorded by an administrator.
decision_reason	TEXT	Required	Reason given for confirming, cancelling, or rescheduling the request.
reviewed_at	TIMESTAMPTZ	Optional	Date and time when an administrator reviewed the record.
reviewed_by	TEXT	Required	Email or identifier of the administrator who reviewed the record.
created_at	TIMESTAMPTZ	Required	Date and time when the record was created.

updated_at	TIMESTAMPTZ	Required	Date and time when the record was last updated.

Table A7 stores museum tour requests, visitor counts, schedules, contact details, and administrative decisions.

Table A8. Data Dictionary for heritage_keeper_events

FIELD NAME	DATA TYPE	CONSTRAINTS	DESCRIPTION
id	BIGINT	PK; Identity; Required	Unique identifier for the record.
user_id	BIGINT	FK → public_users.id; Composite unique; Required	Identifier of the related public user.
source	TEXT	Composite unique; Required	Origin or provenance recorded for the item or event.
item_key	TEXT	Composite unique; Required	Stable application key for the completed or collected item.
completed_at	TIMESTAMPTZ	Required	Date and time when the milestone was completed.

Table A8 records unique learning or exploration milestones completed by a public user.

Table A9. Data Dictionary for heritage_keeper_awards

FIELD NAME	DATA TYPE	CONSTRAINTS	DESCRIPTION
user_id	BIGINT	PK; FK → public_users.id; Required	Identifier of the related public user.
unlocked_at	TIMESTAMPTZ	Required	Date and time when the award was unlocked.

Table A9 records the date on which a public user unlocked the Heritage Keeper award.

Table A10. Data Dictionary for webxr_player_museum_views

FIELD NAME	DATA TYPE	CONSTRAINTS	DESCRIPTION

id	BIGINT	PK; Identity; Required	Unique identifier for the record.
user_id	BIGINT	FK → public_users.id; Composite unique; Required	Identifier of the related public user.
hotspot_key	TEXT	Composite unique; Required	Stable key of the museum hotspot viewed by the user.
viewed_at	TIMESTAMPTZ	Required	Date and time when the hotspot was first viewed.

Table A10 records museum information hotspots viewed by a user in the WebXR experience.

Table A11. Data Dictionary for webxr_player_items

FIELD NAME	DATA TYPE	CONSTRAINTS	DESCRIPTION
id	BIGINT	PK; Identity; Required	Unique identifier for the record.
user_id	BIGINT	FK → public_users.id; Composite unique; Required	Identifier of the related public user.
item_key	TEXT	Composite unique; Required	Stable application key for the completed or collected item.
status	TEXT	Required	Current workflow or publication state of the record.
acquired_scene	TEXT	Required	WebXR scene in which the item was collected.
acquired_at	TIMESTAMPTZ	Required	Date and time when the WebXR item was collected.
deposited_at	TIMESTAMPTZ	Optional	Date and time when the WebXR item was deposited.

Table A11 stores collectible WebXR items acquired or deposited by a user.

Table A12. Data Dictionary for visitor_feedback

FIELD NAME	DATA TYPE	CONSTRAINTS	DESCRIPTION
id	BIGINT	PK; Identity; Required	Unique identifier for the record.
user_id	BIGINT	FK → public_users.id; Optional	Identifier of the related public user.
visit_type	TEXT	Required	Indicates whether the feedback concerns an on-site or website visit.
rating	INTEGER	Required	Visitor satisfaction rating from one to five.
visitor_name	TEXT	Required	Optional name supplied by the visitor.
comments	TEXT	Required	Written feedback supplied by the visitor.
sentiment_label	TEXT	Optional	Predicted sentiment class for the visitor comment.
sentiment_confidence	NUMERIC(5,4)	Optional	Confidence score assigned to the predicted sentiment.
sentiment_scores	JSONB	Required	JSON object containing the sentiment scores returned by the model.
sentiment_model	TEXT	Required	Name or version of the model used for sentiment analysis.
sentiment_provider	TEXT	Required	Service or library that performed sentiment analysis.
sentiment_analyzed_at	TIMESTAMPTZ	Optional	Date and time when sentiment analysis completed.
status	TEXT	Required	Current workflow or publication state of the record.
reviewed_at	TIMESTAMPTZ	Optional	Date and time when an administrator reviewed the record.
reviewed_by	TEXT	Required	Email or identifier of the administrator who reviewed the record.
created_at	TIMESTAMPTZ	Required	Date and time when the record was created.
updated_at	TIMESTAMPTZ	Required	Date and time when the record was last updated.

Table A12 stores visitor ratings, comments, review status, and automated sentiment-analysis results.

Table A13. Data Dictionary for analytics_events

FIELD NAME	DATA TYPE	CONSTRAINTS	DESCRIPTION
id	BIGINT	PK; Identity; Required	Unique identifier for the record.
event_type	TEXT	Required	Application event that generated the notification or analytics record.
module	TEXT	Required	Application module in which the analytics event occurred.
target_id	TEXT	Required	Stable identifier of the page, scene, artifact, or resource involved.
visitor_hash	TEXT	Required	Privacy-preserving visitor identifier used to group anonymous events.
user_id	BIGINT	FK → public_users.id; Optional	Identifier of the related public user.
device_type	TEXT	Required	General device category detected for the visit.
browser_family	TEXT	Required	Browser family detected for the event.
os_family	TEXT	Required	Operating-system family detected for the event.
page_path	TEXT	Required	Application path on which the event occurred.
metadata	JSONB	Required	Bounded JSON object containing event-specific attributes.
created_at	TIMESTAMPTZ	Required	Date and time when the record was created.

Table A13 stores privacy-conscious interaction events used for system usage and engagement reports.

Table A14. Data Dictionary for community_highlights

FIELD NAME	DATA TYPE	CONSTRAINTS	DESCRIPTION
id	BIGINT	PK; Identity; Required	Unique identifier for the record.
title	TEXT	Required	Primary title of the record.
slug	TEXT	Unique; Required	URL-safe unique name used to address the record.
category	TEXT	Required	Classification assigned to the content record.
event_date	DATE	Required	Date associated with the community highlight.
short_description	TEXT	Required	Brief description displayed in highlight listings.
long_description	TEXT	Required	Complete English description of the community highlight.
short_description_tl	TEXT	Required	Brief Tagalog description displayed in highlight listings.
long_description_tl	TEXT	Required	Complete Tagalog description of the community highlight.
read_time_minutes	INTEGER	Optional	Estimated number of minutes required to read the highlight.
image_filename	TEXT	Optional	Stored filename of the primary image.
image_filenames	JSONB	Required	Ordered JSON array of image filenames for the highlight gallery.
status	TEXT	Required	Current workflow or publication state of the record.
is_featured	BOOLEAN	Required	Indicates whether the content is emphasized in featured displays.

is_published	BOOLEAN	Required	Indicates whether the content is publicly visible.
created_at	TIMESTAMPTZ	Required	Date and time when the record was created.
updated_at	TIMESTAMPTZ	Required	Date and time when the record was last updated.

Table A14 stores bilingual community stories, programs, advisories, dates, images, and publication state.

Table A15. Data Dictionary for online_gallery_categories

FIELD NAME	DATA TYPE	CONSTRAINTS	DESCRIPTION
id	BIGINT	PK; Identity; Required	Unique identifier for the record.
slug	TEXT	Unique; Required	URL-safe unique name used to address the record.
display_name	TEXT	Unique; Required	Primary display label of the gallery category.
sort_order	INTEGER	Required	Numeric position used when displaying records in sequence.
is_active	BOOLEAN	Required	Indicates whether the account or category is available for use.
created_at	TIMESTAMPTZ	Required	Date and time when the record was created.
updated_at	TIMESTAMPTZ	Required	Date and time when the record was last updated.
display_name_en	TEXT	Required	English display label of the gallery category.
display_name_tl	TEXT	Required	Tagalog display label of the gallery category.

Table A15 stores ordered bilingual categories used to organize the online gallery.

Table A16. Data Dictionary for online_gallery_items

FIELD NAME	DATA TYPE	CONSTRAINTS	DESCRIPTION

id	BIGINT	PK; Identity; Required	Unique identifier for the record.
category_id	BIGINT	FK → online_gallery_categories.id; Required	Identifier of the gallery category containing the item.
title	TEXT	Required	Primary title of the record.
period	TEXT	Required	Historical period associated with the gallery item.
year_label	TEXT	Required	Display year or date range associated with the content.
source	TEXT	Required	Origin or provenance recorded for the item or event.
location	TEXT	Required	Location associated with the booking or content item.
short_caption	TEXT	Required	Primary short caption retained for compatibility.
full_description	TEXT	Required	Primary full description retained for compatibility.
teaching_note	TEXT	Required	Optional instructional note for educators and learners.
tags	JSONB	Required	JSON array of searchable keywords assigned to the gallery item.
image_filename	TEXT	Optional	Stored filename of the primary image.
model_filename	TEXT	Optional	Stored filename of the optional three-dimensional model.
status	TEXT	Required	Current workflow or publication state of the record.
is_published	BOOLEAN	Required	Indicates whether the content is publicly visible.
sort_order	INTEGER	Required	Numeric position used when displaying records in sequence.
created_at	TIMESTAMPTZ	Required	Date and time when the record was created.
updated_at	TIMESTAMPTZ	Required	Date and time when the record was last updated.
title_tl	TEXT	Required	Tagalog title displayed to visitors.
location_tl	TEXT	Required	Tagalog location label displayed to visitors.

short_caption_tl	TEXT	Required	Tagalog short caption displayed in gallery listings.
full_description_tl	TEXT	Required	Complete Tagalog description of the gallery item.
title_en	TEXT	Required	English title displayed to visitors.
location_en	TEXT	Required	English location label displayed to visitors.
short_caption_en	TEXT	Required	English short caption displayed in gallery listings.
full_description_en	TEXT	Required	Complete English description of the gallery item.

Table A16 stores bilingual artifact records, descriptive metadata, media references, and publication state.

Table A17. Data Dictionary for edu_resources

FIELD NAME	DATA TYPE	CONSTRAINTS	DESCRIPTION
id	BIGINT	PK; Identity; Required	Unique identifier for the record.
title	TEXT	Required	Primary title of the record.
slug	TEXT	Unique; Required	URL-safe unique name used to address the record.
category	TEXT	Required	Classification assigned to the content record.
educator_name	TEXT	Required	Name of the educator who prepared or contributed the resource.
subject_topic	TEXT	Required	Subject area or learning topic associated with the resource.
author	TEXT	Required	Named author of the educational material.
year_label	TEXT	Required	Display year or date range associated with the content.

description	TEXT	Required	Full description of the content or virtual-tour environment.
file_filename	TEXT	Optional	Stored filename of the downloadable educational resource.
cover_image_filename	TEXT	Optional	Stored filename of the resource cover image.
status	TEXT	Required	Current workflow or publication state of the record.
is_published	BOOLEAN	Required	Indicates whether the content is publicly visible.
created_at	TIMESTAMPTZ	Required	Date and time when the record was created.
updated_at	TIMESTAMPTZ	Required	Date and time when the record was last updated.

Table A17 stores educational resources, descriptive metadata, downloadable files, and publication state.

Table A18. Data Dictionary for community_participation_submissions

FIELD NAME	DATA TYPE	CONSTRAINTS	DESCRIPTION
id	BIGINT	PK; Identity; Required	Unique identifier for the record.
user_id	BIGINT	FK → public_users.id; Optional	Identifier of the related public user.
submission_type	TEXT	Required	Identifies the submission as a teacher resource or community story.
title	TEXT	Required	Primary title of the record.
resource_category	TEXT	Required	File or resource category selected by the contributor.
story_category	TEXT	Required	Community-story category selected by the contributor.
submitter_name	TEXT	Required	Name supplied by the person submitting the material.
submitter_email	TEXT	Required	Email address used to verify and contact the contributor.
subject_topic	TEXT	Required	Subject area or learning topic associated with the resource.
description	TEXT	Required	Full description of the content or virtual-tour environment.

story_content	TEXT	Required	Full written content of the submitted community story.
resource_file_filename	TEXT	Optional	Stored filename of the submitted educational resource.
photo_filename	TEXT	Optional	Stored filename of the optional submission photograph.
is_private	BOOLEAN	Required	Indicates whether the contributor requested private handling.
status	TEXT	Required	Current workflow or publication state of the record.
verification_code_hash	TEXT	Optional	Hash of the code used to verify the contributor's email.
verification_sent_at	TIMESTAMPTZ	Optional	Date and time when the verification code was sent.
verification_expires_at	TIMESTAMPTZ	Optional	Expiration date and time of the verification code.
verified_at	TIMESTAMPTZ	Optional	Date and time when the submission email was verified.
reviewed_at	TIMESTAMPTZ	Optional	Date and time when an administrator reviewed the record.
reviewed_by	TEXT	Required	Email or identifier of the administrator who reviewed the record.
created_at	TIMESTAMPTZ	Required	Date and time when the record was created.
updated_at	TIMESTAMPTZ	Required	Date and time when the record was last updated.

Table A18 stores teacher resources and community stories submitted for verification and administrative review.

Table A19. Data Dictionary for virtual_tours

FIELD NAME	DATA TYPE	CONSTRAINTS	DESCRIPTION
id	BIGINT	PK; Identity; Required	Unique identifier for the record.
slug	TEXT	Unique; Required	URL-safe unique name used to address the record.
title_en	TEXT	Required	English title displayed to visitors.
title_tl	TEXT	Required	Tagalog title displayed to visitors.
description_en	TEXT	Required	English value stored for description.
description_tl	TEXT	Required	Tagalog value stored for description.
captured_on	DATE	Optional	Capture date retained for compatibility; current

			scene captures are stored per scene.
status	TEXT	Required	Current workflow or publication state of the record.
is_system_original	BOOLEAN	Required	Indicates whether the record represents the protected packaged original.
published_revision_id	BIGINT	FK → virtual_tour_revisions.id; Optional	Identifier of the revision currently served to visitors.
draft_revision_id	BIGINT	FK → virtual_tour_revisions.id; Optional	Identifier of the working revision currently edited by administrators.
sort_order	INTEGER	Required	Numeric position used when displaying records in sequence.
created_by	TEXT	Required	Email or identifier of the administrator who created the record.
published_at	TIMESTAMPTZ	Optional	Date and time when the revision or content was published.
created_at	TIMESTAMPTZ	Required	Date and time when the record was created.
updated_at	TIMESTAMPTZ	Required	Date and time when the record was last updated.

Table A19 stores the identity, bilingual metadata, publication pointers, and ordering of each 360-degree tour environment.

Table A20. Data Dictionary for virtual_tour_revisions

FIELD NAME	DATA TYPE	CONSTRAINTS	DESCRIPTION
id	BIGINT	PK; Identity; Required	Unique identifier for the record.
tour_id	BIGINT	FK → virtual_tours.id; Composite unique; Required	Identifier of the virtual-tour environment owning the revision.
version_number	INTEGER	Composite unique; Required	Sequential revision number within the parent tour or site map.
state	TEXT	Required	Lifecycle state of the revision.
created_by	TEXT	Required	Email or identifier of the administrator who created the record.

published_at	TIMESTAMP TZ	Optional	Date and time when the revision or content was published.
created_at	TIMESTAMP TZ	Required	Date and time when the record was created.
updated_at	TIMESTAMP TZ	Required	Date and time when the record was last updated.
map_config	JSONB	Required	JSON object configuring the upper- right minimap and its scene-linked markers.

Table A20 stores immutable publication versions and editable drafts of a virtual tour, including minimap configuration.

Table A21. Data Dictionary for virtual_tour_scenes

FIELD NAME	DATA TYPE	CONSTRAINTS	DESCRIPTION
id	BIGINT	PK; Identity; Required	Unique identifier for the record.
revision_id	BIGINT	FK → virtual_tour_revisions .id; Composite unique; Required	Identifier of the virtual-tour revision owning the scene or hotspot.
scene_key	TEXT	Composite unique; Required	URL-safe stable key used to address the 360-degree scene.
name_en	TEXT	Required	English name displayed for the scene or hotspot.
name_tl	TEXT	Required	Tagalog name displayed for the scene or hotspot.
captured_on	DATE	Required	Capture date retained for compatibility; current scene captures are stored per scene.
panorama_path	TEXT	Required	Storage path of the equirectangular panorama image.
panorama_width	INTEGER	Required	Pixel width recorded for panorama validation.
panorama_height	INTEGER	Required	Pixel height recorded for panorama validation.

panorama_mime	TEXT	Required	Validated MIME type of the panorama image.
validation_status	TEXT	Required	Result of the panorama format and dimension validation.
initial_pitch	NUMERIC(7, 2)	Required	Initial vertical viewing angle of the scene, in degrees.
initial_yaw	NUMERIC(7, 2)	Required	Initial horizontal viewing angle of the scene, in degrees.
initial_hfov	NUMERIC(6, 2)	Required	Initial horizontal field of view of the scene, in degrees.
is_start_scene	BOOLEAN	Required	Indicates whether visitors enter the tour through this scene.
sort_order	INTEGER	Required	Numeric position used when displaying records in sequence.
created_at	TIMESTAMPTZ	Required	Date and time when the record was created.
updated_at	TIMESTAMPTZ	Required	Date and time when the record was last updated.

Table A21 stores each validated equirectangular panorama and its initial 360-degree viewing orientation.

Table A22. Data Dictionary for virtual_tour_hotspots

FIELD NAME	DATA TYPE	CONSTRAINTS	DESCRIPTION
id	BIGINT	PK; Identity; Required	Unique identifier for the record.
revision_id	BIGINT	FK → virtual_tour_revisions.id; Required	Identifier of the virtual-tour revision owning the scene or hotspot.

scene_id	BIGINT	FK → virtual_tour_scenes.id; Required	Identifier of the virtual-tour scene containing the hotspot.
hotspot_type	TEXT	Required	Identifies the hotspot as an information item or scene navigation control.
name_en	TEXT	Required	English name displayed for the scene or hotspot.
name_tl	TEXT	Required	Tagalog name displayed for the scene or hotspot.
pitch	NUMERIC(7,2)	Required	Vertical hotspot position in the 360- degree scene, in degrees.
yaw	NUMERIC(7,2)	Required	Horizontal hotspot position in the 360-degree scene, in degrees.
target_scene_id	BIGINT	FK → virtual_tour_scenes.id; Optional	Identifier of the destination scene for a navigation hotspot.
target_pitch	NUMERIC(7,2)	Optional	Vertical viewing angle used after entering the destination scene.
target_yaw	NUMERIC(7,2)	Optional	Horizontal viewing angle used after entering the destination scene.
target_hfov	NUMERIC(6,2)	Optional	Horizontal field of view used after entering the destination scene.
image_path	TEXT	Optional	Storage path of the optional hotspot image.
glb_path	TEXT	Optional	Storage path of the optional GLB three-dimensional model.
subtitle_en	TEXT	Required	English narration or information text for the hotspot.
subtitle_tl	TEXT	Required	Tagalog narration or information text for the hotspot.

audio_en_path	TEXT	Optional	Storage path of the optional English hotspot narration.
audio_tl_path	TEXT	Optional	Storage path of the optional Tagalog hotspot narration.
tts_voice_id	TEXT	Required	Configured text-to-speech voice identifier.
tts_model_id	TEXT	Required	Configured text-to-speech model identifier.
nested_hotspots	JSONB	Required	JSON array of secondary hotspots associated with the item.
is_enabled	BOOLEAN	Required	Indicates whether the hotspot is available to visitors.
sort_order	INTEGER	Required	Numeric position used when displaying records in sequence.
created_at	TIMESTAMP TZ	Required	Date and time when the record was created.
updated_at	TIMESTAMP TZ	Required	Date and time when the record was last updated.

Table A22 stores item and navigation hotspots positioned inside a virtual-tour scene, including bilingual media and target views.

Table A23. Data Dictionary for site_maps

FIELD NAME	DATA TYPE	CONSTRAINTS	DESCRIPTION
id	BIGINT	PK; Identity; Required	Unique identifier for the record.
is_system_original	BOOLEAN	Required	Indicates whether the record represents the protected packaged original.
published_revision_id	BIGINT	FK → site_map_revisions.id; Optional	Identifier of the revision currently served to visitors.
draft_revision_id	BIGINT	FK → site_map_revisions.id; Optional	Identifier of the working revision currently edited by administrators.
created_by	TEXT	Required	Email or identifier of the administrator who created the record.
published_at	TIMESTAMPTZ	Optional	Date and time when the revision or content was published.
created_at	TIMESTAMPTZ	Required	Date and time when the record was created.
updated_at	TIMESTAMPTZ	Required	Date and time when the record was last updated.

Table A23 stores the protected site-map record and pointers to its current published and working revisions.

Table A24. Data Dictionary for site_map_revisions

FIELD NAME	DATA TYPE	CONSTRAINTS	DESCRIPTION
id	BIGINT	PK; Identity; Required	Unique identifier for the record.
site_map_id	BIGINT	FK → site_maps.id; Composite unique; Required	Identifier of the site-map record owning the revision.
version_number	INTEGER	Composite unique; Required	Sequential revision number within the parent tour or site map.
state	TEXT	Required	Lifecycle state of the revision.
config	JSONB	Required	JSON object containing site-map views, coordinates, hotspot content, media, links, and intro audio.
created_by	TEXT	Required	Email or identifier of the administrator who created the record.
published_at	TIMESTAMPTZ	Optional	Date and time when the revision or content was published.
created_at	TIMESTAMPTZ	Required	Date and time when the record was created.
updated_at	TIMESTAMPTZ	Required	Date and time when the record was last updated.

Table A24 stores revisioned site-map configuration, including aerial/360 media, hotspots, links, and narration settings as JSON.